

Rajshahi University of Engineering & Technology, Rajshahi-6204



Course No.: MTE 3202

Course Title: Power Electronics and Drives Sessional

Project Report

Project Title

Design and Implementation of a DC-DC Boost Converter (9V to 24V)
using ESP32-based PWM Control

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Chapter 1: Power Electronics Knowledge

1.1 Introduction:

Efficiency and power density have been at the heart of the power electronics community over the course of the last five decades. The improvements in efficiency and power density are heavily driven by power technologies, circuit topologies, and control methods. The historical development of power electronics, and reliability engineering. With the increasing demands of application-driven research, reliability has become a considerable practical challenge in power electronics. Reliability is an important performance factor that is considered during the design, manufacturing, and field operation of power electronic converters. Over the past decade, the business model in the industry has been transitioning from product suppliers to holistic service providers. This demands life-cycle-cost reduction and operation optimization of power electronic converters through innovative design, in-depth understanding of failure mechanisms and mission profiles, and predictive maintenance [1]. This work presents a classification of the main power quality problems and the respective context with the standards, a review of power quality problems related to the power production from renewables, the contextualization with electric mobility and electrical systems, a power electronics solutions to compensate the main power quality problems, as well as power electronics solutions to guarantee high levels of power quality. Relevant experimental results and exemplificative developed power electronics prototypes are also presented throughout the work [2].

1.2 DC-DC Boost Converter:

Direct current (DC to DC) converters are circuits which convert sources of direct current (DC) from one voltage level to another by changing the duty cycle of the main switches in the circuits. Since DC-DC converters are nonlinear systems, they represent a big challenge for control design [3]. DC converters are used in DC voltage regulators; and also, are used, with an inductor in conjunction, to generate a DC current source, specifically for the current source inverter [4]. It works by storing energy in the inductor when the switch is closed and releasing it to the load when the switch is open.

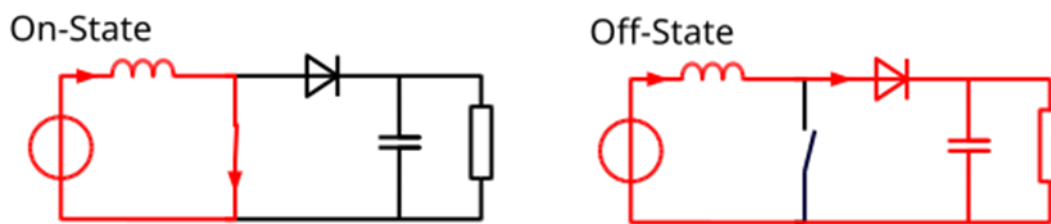


Fig.1. ON and OFF states of DC/DC Boost Converter

The relation between input voltage, output voltage, and duty cycle is given as:

$$V_o = \frac{V_{in}}{1 - D}$$

Where:

V_o = Output voltage

V_{in} = Input voltage

D = Duty cycle

For this design, the theoretical duty cycle was calculated as:

$$D = 1 - \frac{9}{24} = 0.625 \text{ (62.5\%)}$$

However, in the experimental setup, a 47.5% duty cycle was used to obtain around 24V due to practical non-idealities.

Since classical control methods are designed at one nominal operating point, they are not able to respond satisfactorily to operating point variations and load disturbance. They often fail to perform satisfactorily under large parameter or load variations [3].

The boost type DC-DC converters are used in applications where the required output voltage needed to be higher than the source voltage. The control of this type DC-DC converters are more difficult than the buck type where the output voltage is smaller than the source voltage. The difficulties in the control of boost converters are due to the non-minimum phase structure since, the control input appears both in voltage and current equations, from the control point of view the control of boost type converters are more difficult than buck type.

1.3 PWM DC to DC converter:

PWM DC-to-DC converters are very popular for the last three decades, and that are widely used at all power levels. Since switching converters constitute a case of variable structure systems, the conventional control technique can be a possible option to control this kind of circuits [5]. When designing dc-dc power converters using pulse width modulation (PWM), fault analysis is crucial. It provides us with comprehensive information about the potential harm and aids in our understanding of the severity of different fault types. It provides us with precise details regarding the highest resist durability of each part of a converter. It is quite helpful to us while constructing the converters to operate within their maximum capabilities. In this section we measure the converter's performance after introducing various flaws [6].

1.4 ESP-32 based Control:

ESP32 is a powerful and cost-effective platform for developing electronics applications. ESP32 has the following features: (1) a dual-core processor, (2) a large number of general-purpose input/output (GPIO) pins, and (3) low power consumption, which provides higher processing power and facilitates multitasking and efficient execution of complex tasks. It gives the user precise control over frequency, amplitude, and waveform shape, and it is usually used for testing and stimulus purposes. In addition, Analog Discovery offers two programmable power supplies, with adjustable voltage outputs for powering external circuits or components. The device also has a built-in logic analyzer that captures and analyzes digital signals with up to 32 channels [7]. The ESP32 produces PWM (Pulse Width Modulation) signals using its built-in timers to manage the MOSFET switching in the boost converter. By varying the PWM duty cycle, it controls the output voltage in relation to the input.

Chapter 2: Designing Engineering Solutions

2.1 Objectives

1. To design and implement a DC-DC boost converter capable of converting 9V input to approximately 24V output.
2. To utilize ESP32 for generating PWM signals for converter control.
3. To validate the design through MATLAB simulation and practical hardware implementation.

2.2 Methodology:

Theoretical Design

- a. The duty cycle for converting 9V input to 24V output was calculated.
- b. Component values such as inductor size, capacitor rating, and transistor type were selected based on the design requirements.

Simulation in MATLAB

- a. The boost converter circuit was modeled with inductor, transistor, diode, and capacitor.
- b. A 9V input and desired 24V output were set.
- c. Duty cycles of 47.5% and 62.5% were tested to observe voltage boosting and ripple performance.

PWM Implementation on ESP32

- a. ESP32 was programmed to generate a 100 kHz PWM signal with 47.5% duty cycle on GPIO21.
- b. The PWM signal was verified using an oscilloscope in Simulation before connecting it to the circuit.

Hardware Construction:

- a. The circuit was built on a breadboard following the boost converter topology.
- b. ESP32 output was connected to the TIP120 transistor, which switched the inductor current.
- c. Diode and capacitor were added to complete the circuit.

Testing and Measurement:

- a. A 9V battery was applied as the input source.
- b. The output voltage was measured across the load resistor using a digital multimeter.
- c. The measured output was compared with MATLAB simulation results for validation

2.3 Component List:

Component Name	Functionality	Price (BDT)
ESP32 Development Board	Generates PWM control signal	449
Inductor (680 μ H)	Stores and releases magnetic energy	10
Capacitor (100 nF)	Smooths output voltage ripple	4
TIP120 Transistor	Main switching device	25
Fast Recovery Diode (UF4007)	Provides current path during OFF cycle	5
Resistor (10 k Ω)	Load/stabilization resistor	1
DC Source (9 V battery)	Input supply	75
Load Resistor (10 k Ω)	For output voltage testing	1
Total Price		= 570 BDT

2.4 Design and Simulation:

The converter was designed in MATLAB/Simulink with 9V input, 24V target, 680 μ H inductor, 100 nF capacitor, transistor, diode and 10 k Ω load resistor. Simulation waveforms showed inductor current buildup and capacitor charging. The output exceeded 24V at 62.5% duty cycle, while 47.5% duty cycle gave a more stable 24V output.

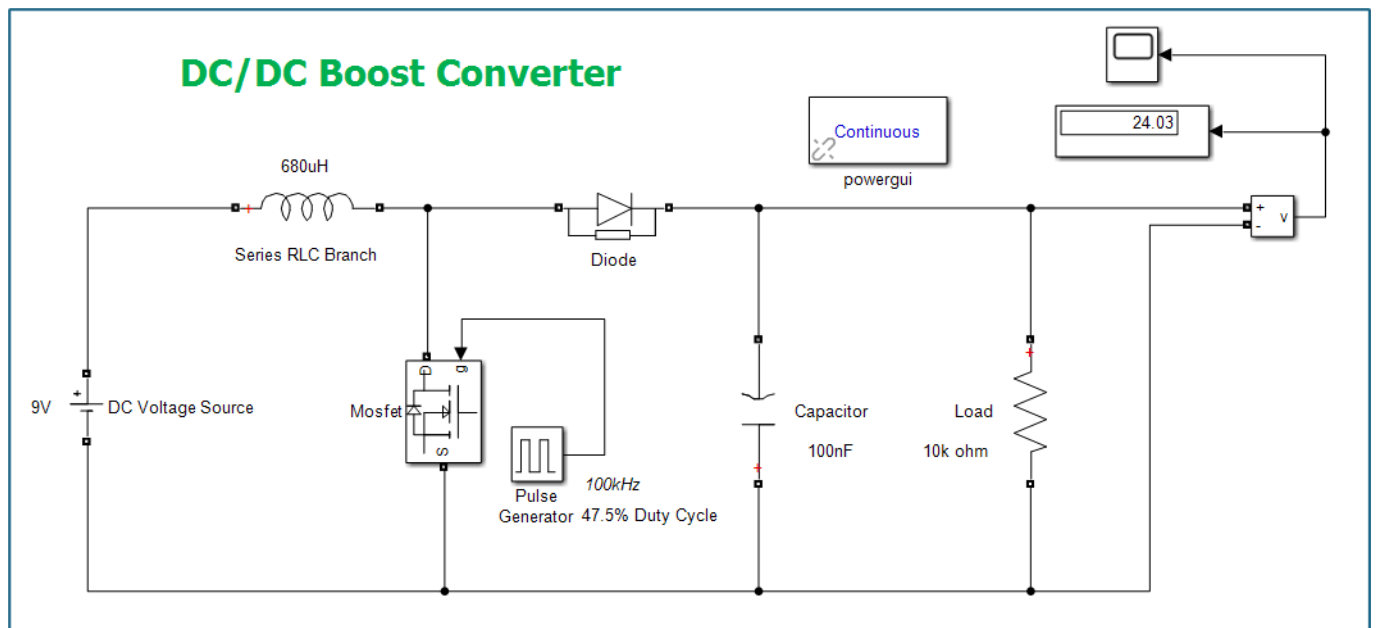


Fig. 2. Simulink Model of the DC/DC Boost Converter (9v to 24v)

2.5 PWM Generation with ESP32:

Pulse Width Modulation (PWM) is a technique used to control the effective voltage by switching a signal between HIGH and LOW states. The average voltage depends on the duty cycle, which is the ratio of ON time to the total switching period.

The ESP32 microcontroller has built-in hardware PWM functionality through the LEDC (LED Control) module. It allows precise control of frequency, duty cycle, and resolution across multiple channels.

In this project, the ESP32 was programmed to generate a PWM signal at 100 kHz with a 47.5% duty cycle. The PWM signal controlled the TIP120 transistor, which switched the inductor current, enabling the boost converter to step up the voltage from 9V to ~24V

2.6 Simulation Result:

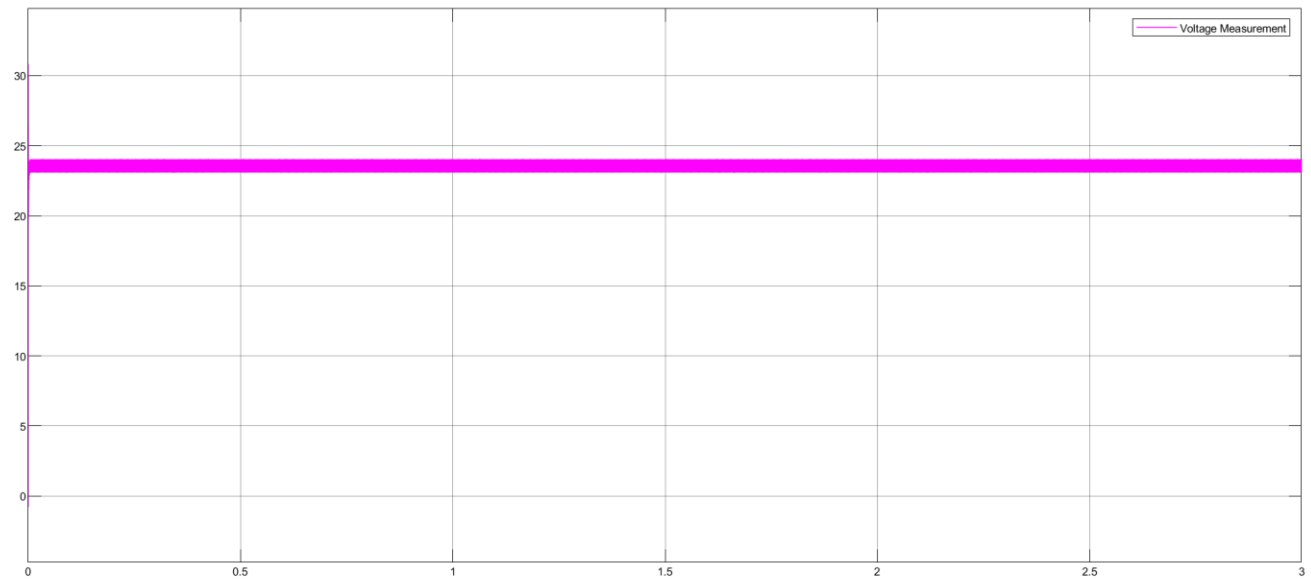


Fig. 3. Voltage output across the resistor during Simulation with 47.5% Duty Cycle

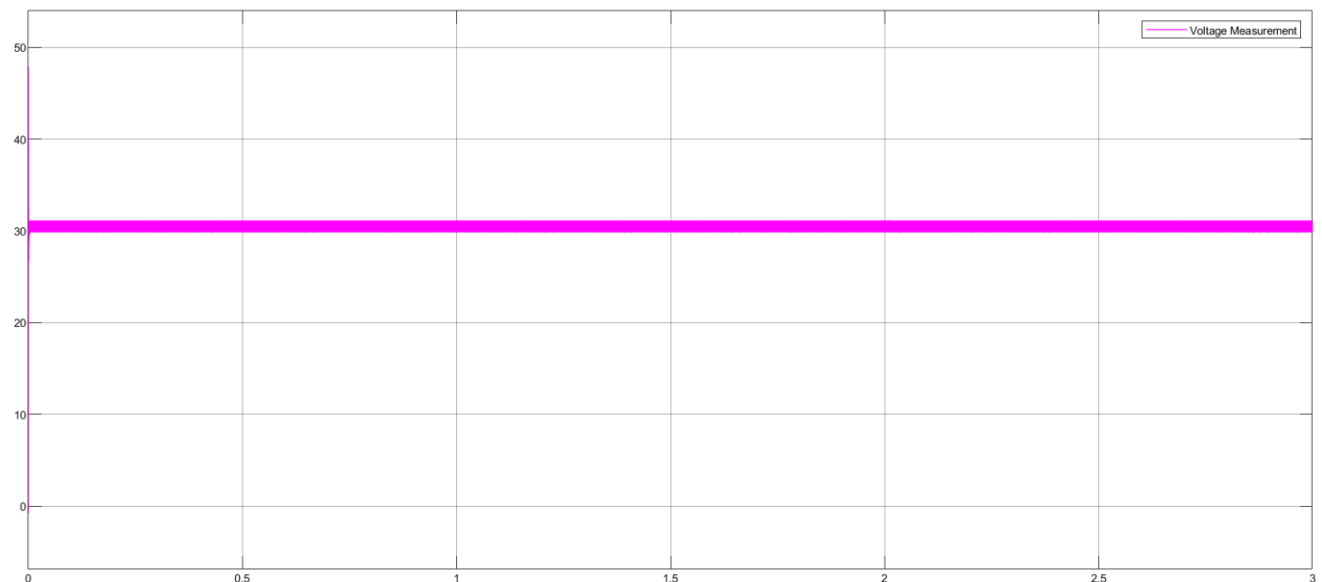


Fig. 4. Voltage output across the resistor during Simulation with 62.5% Duty Cycle

2.7 Calculations:

Duty Cycle (Theoretical):

$$V_o = \frac{V_{in}}{1 - D}$$

$$D = 1 - \frac{9}{24} = 0.625 \text{ (62.5\%)}$$

In practical implementation, a **47.5% duty cycle** was used, which provided an output voltage close to 24V due to non-idealities in the components.

Inductor Selection:

The inductor value was chosen to limit current ripple and ensure continuous conduction. The inductor current ripple is given by:

$$\Delta I_L = \frac{V_{in} \times D}{L \times f_s}$$

Substituting the values:

$$\Delta I_L = \frac{9 \times 0.5}{680 \times 10^{-6} \times 100 \times 10^3} \approx 0.066 \text{ A}$$

Thus, a **680 μ H inductor** was selected to maintain manageable current ripple.

Capacitor Selection:

The output capacitor was chosen to reduce voltage ripple. The approximate voltage ripple is given by:

$$\Delta V_o = \frac{I_o \times D}{C \times f_s}$$

Assuming a load current $I_o = 0.1 \text{ A}$:

$$\Delta V_o = \frac{0.1 \times 0.5}{100 \times 10^{-9} \times 100 \times 10^3} \approx 5 \text{ V}$$

A **100 nF capacitor** was used in this experiment, but a larger capacitor would further reduce the ripple for more stable output.

Switching Frequency:

$$f_s = 100 \text{ kHz}$$

High frequency was selected to reduce inductor and capacitor size while maintaining efficiency.

2.8 Hardware Implementation:

The circuit was built on a breadboard using ESP32, TIP120 transistor, inductor, capacitor, and diode. ESP32 generated a **100 kHz PWM with 47.5% duty cycle** at GPIO21, which switched the transistor. When 9V input was applied, the converter successfully boosted the voltage to ~24V.

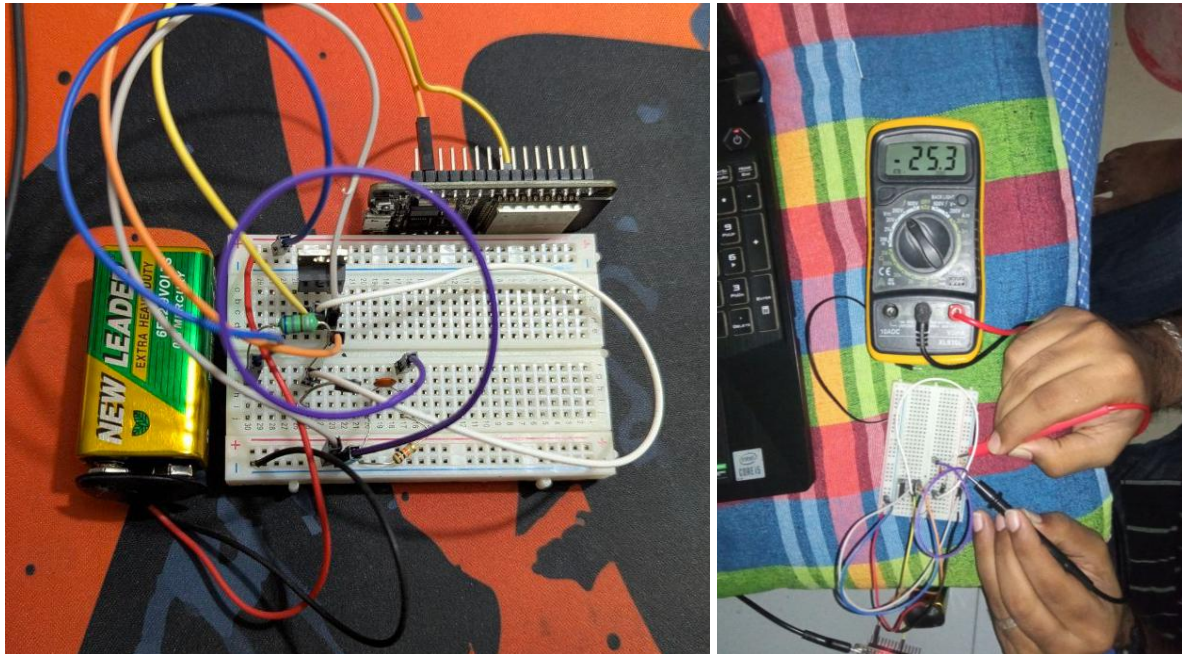


Fig. 5. Hardware Implementation of the DC/DC Boost Converter and Testing

2.9 ESP32 PWM Code Used:

```
// ESP32 Boost Converter PWM, Vin = 9V, Vout = 24V, Duty Cycle ≈ 47.5%, Freq = 100kHz

const int pwmPin = 21;           // GPIO21 output
const int pwmChannel = 0;        // PWM channel
const int pwmFreq = 100000;      // 100 kHz switching frequency
const int pwmResolution = 8;     // 8-bit resolution (0-255)
// Duty cycle for 47.5% (24V out from 9V in)
int dutyCycle = 122; // ≈ 47.5%

void setup() { // Setup PWM
    ledcSetup(pwmChannel, pwmFreq, pwmResolution);
    ledcAttachPin(pwmPin, pwmChannel);
    ledcWrite(pwmChannel, dutyCycle); // Apply duty cycle
}

void loop() {
    // Constant 100 kHz, 47.5% duty }
```

Chapter 3: Contextual Reasoning in Power Electronics

Power electronics is not only the science of switching devices, voltage conversion, or current regulation, it is also an art of reasoning in context. When an engineer works with circuits such as a DC-DC boost converter, the process involves more than solving equations; it involves understanding the real-world context, anticipating problems, making trade-offs, and drawing meaningful conclusions from both theory and practice.

3.1 Why Context Matters in Power Electronics

In textbooks, we see neat equations such as:

$$V_o = \frac{V_{in}}{1 - D}$$

where V_o is the output voltage, V_{in} is the input voltage, and D is the duty cycle. Theoretically, if we want to step up 9V to 24V, the equation tells us to use a 62.5% duty cycle.

But when the circuit was built in the lab, something different happened; the converter produced almost 24V at just 47.5% duty cycle.

There are resistive losses, switching delays, and diode drops that shift the behaviour of the system. This gap between theory and practice is exactly where contextual reasoning becomes essential. An engineer must learn to interpret, adapt, and refine designs based on the environment in which they are applied.

3.2 Reasoning Behind Design Decisions

Every design choice in this project was guided by contextual reasoning:

Microcontroller (ESP32): Selected not only because it could generate PWM signals, but also because it is affordable, widely available, and easy to program. In a developing country or a resource-limited lab, these factors matter as much as raw technical performance.

Inductor (680 μ H): The choice of inductor was not random. A smaller inductor would have made the ripple intolerable; a much larger one would be expensive and bulky. Thus, the 680 μ H value reflects a balance between practicality and theoretical requirements. Also, it was verified by the simulations as well.

Switching Frequency (100 kHz): Too low a frequency would need huge inductors and capacitors. Too high a frequency would create excessive switching losses. Choosing 100 kHz was again a reasoning exercise balancing efficiency, component size, and feasibility.

Capacitor (100 nF): Though small, it demonstrated the concept effectively within budget limits. The reasoning here was educational value over absolute performance.

This shows that engineering is not about finding a "perfect" solution, but about finding a workable, contextual solution.

3.3 Bridging Simulation and Experiment

The project used MATLAB/Simulink simulations before building the hardware.

In simulation: The circuit behaved in a nearly ideal way, producing smooth waveforms and stable voltage.

In hardware: Results included noise, ripple, and unexpected shifts in duty cycle requirements.

Instead of treating these differences as "errors," we treated them as lessons in contextual reasoning. We realized that simulations provide guidance, but experiments reveal the truth of the real world. The act of comparing both and reasoning through the differences is one of the most valuable skills an engineer can develop.

3.4 Economic, Educational, and Social Context

Contextual reasoning does not stop at the lab bench it also considers the wider context:

Economic: The project cost was only 570 BDT, making it accessible to students. This shows that innovation does not always require high budgets.

Educational: We learned not only circuit theory but also the discipline of adapting to limitations skills that prepare them for real engineering challenges.

Social: In countries striving for energy independence, small-scale, low-cost power electronics solutions like this can play a crucial role in rural electrification, off-grid applications, and renewable integration.

Thus, the reasoning behind every choice connects theory, practice, economy, and society.

3.5 Conclusion

Contextual reasoning in power electronics is what makes designs successful outside the classroom. Equations may describe what “should” happen, but reasoning explains why something else “does” happen and how to adapt accordingly. The boost converter project illustrates this journey beautifully: from theory to practice, from simulation to hardware, from ideal assumptions to real-world compromises.

Chapter 4: Sustainable Applications and Environmental Impact

In recent days, as engineers we cannot ignore sustainability. Power electronics, especially converters like the one built in this project, are at the heart of the global shift toward clean energy and efficient systems. A simple 9V to 24V boost converter may appear small, but its principle underpins the technologies that power solar homes, electric vehicles, and LED lighting.

4.1 Role of Boost Converters in Sustainable Technology

Boost converters play critical roles in many green applications:

Solar Energy: Solar panels often generate only 9-18V, which is too low for charging batteries or running appliances. A boost converter raises this voltage to a stable, usable level.

Wind Turbines: Small turbines produce fluctuating outputs; converters stabilize them for storage.

Battery Devices: From medical equipment to mobile gadgets, boost converters extend battery life by squeezing usable energy from low-voltage sources.

Electric Vehicles: E-bikes and EVs rely on converters for efficient battery-to-motor energy transfer, reducing fossil-fuel dependence.

LED Lighting: LEDs need regulated higher DC voltages. Boost converters supply this efficiently, saving both power and costs.

Through these roles, a boost converter is not just a circuit it is a building block of sustainability.

4.2 Environmental Benefits

Using efficient converters brings several environmental advantages:

- 1. Lower Carbon Footprint:** Efficient conversion means less wasted energy, lowering CO₂ emissions.
- 2. Resource Conservation:** By prolonging battery life and improving energy usage, converters reduce the need for frequent replacements, lowering electronic waste.
- 3. Scalability to Green Grids:** The same principles can be scaled up to megawatt solar or wind farms, ensuring that renewable sources integrate smoothly into power systems.

4.3 Hidden Environmental Costs

Sustainability also requires honesty about challenges:

Component Waste: Cheap parts may degrade faster, generating more e-waste.

Efficiency Limits: Using a TIP120 transistor instead of a MOSFET increases energy loss as heat. While affordable, it reduces long-term efficiency.

Material Extraction: Capacitors, inductors, and semiconductors depend on mined resources, which carry their own ecological impact.

Thus, the sustainability of a design must consider not only its benefits but also its costs to the environment.

4.4 Future Pathways to Greener Design

This project could be improved to enhance its sustainability:

MOSFET Instead of TIP120: This change would drastically improve efficiency.

Larger Output Capacitor: Would reduce voltage ripple, prolonging device and battery life.

IoT-Enabled ESP32 Control: Adaptive duty cycle algorithms could adjust performance dynamically, minimizing waste.

Recyclable Materials: Using recyclable circuit boards and components would further reduce long-term impact.

These steps show how even a small academic project can evolve into a truly green design.

4.5 Educational Sustainability

Finally, one of the biggest contributions of this project is educational sustainability. By learning how converters work, students are not only building circuits—they are preparing to build the future. Each project like this creates engineers who can design greener devices, optimize renewable systems, and reduce environmental harm. In this way, education itself becomes a sustainable resource for society.

4.6 Conclusion

The boost converter project demonstrates that sustainability is not an abstract idea it is present in every circuit we build. Whether in solar panels powering a village or in an LED lamp reducing electricity bills, converters help create a cleaner world. By considering efficiency, component selection, and future improvements, engineers ensure that their designs serve both humanity and the planet.

Chapter 5: Effective Individual and Team Performance

Engineering projects are collective journeys where success depends not only on technical knowledge but also on how effectively individuals and teams perform. The development of the DC-DC boost converter (9V to 24V) using ESP32 PWM control highlighted the importance of both individual excellence and teamwork. This chapter reflects on each member's contributions, how the group coordinated tasks, and the skills developed along the way.

5.1 Individual Contributions

Each student took responsibility for specific tasks, ensuring that the workload was balanced and that every stage of the project received full attention. Contributions can be summarized according to roll numbers:

Roll	Contribution	Details
2008011	Coding	Took primary responsibility for programming the ESP32 microcontroller. Developed and refined PWM code to achieve a 100 kHz frequency with the correct duty cycle for smooth converter operation.
2008012	Report Writing	Played a key role in preparing the written report, ensuring clarity and structure. Contributed to documenting project details, results, and insights for the final submission.
2008013	Circuit Build-up	Concentrated on assembling the hardware, connecting components like the TIP120 transistor, inductor, diode, and capacitor, ensuring wiring integrity and stability during testing.
2008014	Report Writing	Focused on preparing the written report, refining technical language, formatting results, and ensuring clarity and coherence in the documentation.
2008015	Calculations	Performed theoretical calculations, including duty cycle determination, inductor current ripple, and capacitor sizing, ensuring accuracy for both simulation and practical setups.
2008016	Coding & Simulations	Contributed to programming the ESP32 microcontroller and modelled the boost converter circuit in MATLAB/Simulink. Tested multiple duty cycles and analysed voltage ripple to validate theoretical predictions.
2008017	Calculations	Assisted in theoretical analysis, performing calculations for duty cycle, ripple values, and capacitor sizing to support both simulation and hardware implementation.
2008018	Circuit Build-up	Actively worked on constructing and testing the hardware setup, troubleshooting component issues to ensure reliable performance during practical implementation.
2008019	Simulations and Circuit Testing	Modelled the boost converter in MATLAB/Simulink, analysed input-output relationships and waveform behaviours, and assisted in testing the physical circuit to verify performance.
2008020	Report Writing	Assisted in preparing detailed documentation, with a special focus on Chapter 5 (Teamwork and Individual Performance), providing reflective insights on collaboration and personal learning.

5.2 Team Coordination

Although roles were divided, the team worked in a highly collaborative manner. Simulation results were continuously shared with the coding group to fine-tune PWM generation. Similarly, theoretical calculations were verified against both MATLAB models and experimental hardware. Report writers collected data and experiences from all groups to create a cohesive and professional document.

5.3 Overcoming Challenges Together

The team faced several obstacles, including mismatches between theoretical duty cycles (62.5%) and practical implementation (47.5%), unexpected output ripple, and voltage drops due to non-ideal components. These challenges were not solved individually but collectively by combining coding expertise, hardware adjustments, and simulation analysis.

5.4 Skill Development

Through this collaborative work, the team developed both technical and soft skills:

Technical Skills: Coding microcontrollers, running simulations, circuit building, and analytical calculations.

Soft Skills: Communication, time management, problem-solving, and collaborative writing.

Every member left the project with new confidence in both engineering and teamwork.

5.5 Conclusion

The success of the DC-DC boost converter project was not the result of one person's effort, but of a well-coordinated team where everyone played their part. By distributing roles according to strengths and interests, the group achieved a balance between individual excellence and collective achievement. This chapter itself stands as proof of that teamwork, documenting the contributions of each member and celebrating how effective collaboration transforms technical challenges into shared success.

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